



Department of Education

STEM Education
National Schools of Excellence
Biology Teachers Resource Book
Grades 11
Term 3

DOMAIN PHYLA

Science **Biology**

Technology

Engineering

Mathematics

Issued free to schools by the Department of Education

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Acknowledgements

This STEM Biology Teacher Resource Book for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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Contents

SECRETARY'S MESSAGE	5
INTRODUCTION	6
RATIONALE	8
NEW NOMENCLATURE.....	9
STEM Domains.....	10
STEM BIOLOGY CONTENT OVERVIEW – TERM 3.....	11
Table 1.1: Thema Sequence and Duration	11
Table 1.2 Sequencing & Benchmarking STEM Biology Content Standards	11
Table 1.3 Mapping Content Progressions by Modules.....	12
THEMA: ENERGY	13
MODULE: BIOLOGICAL ENERGY SYSTEMS.....	13
Linking Previous Module	13
Module Broad Content Standard.....	13
Module Content Benchmark.....	13
Topic 1: General Energy Types.....	13
Performance Indicators	13
Content Expansion	14
Topic 2: Energy in Animal Cells	14
Performance Indicators	14
Content Expansion	14
Topic 3: Energy in Plant Cells	15
Performance Indicators	15
Content Expansion	15
Topic 4: Energy and Environment.....	15
Performance Indicators	15
Content Expansion	15
Application of Knowledge, Skills and Values (AKSA)	16
THEMA: BATTERY TECHNOLOGY	17
MODULE: BIOLOGICAL BATTERY SYSTEMS.....	17

Linking Previous Module	17
Module Broad Content Standard	17
Module Content Benchmark	17
Topic 1: Concepts of Battery	17
Performance Indicators	17
Content Expansion	18
Topic 2: Biological Inspirations of Battery Technology.....	18
Performance Indicators	18
Content Expansion	18
Topic 3: Bio-Batteries.....	19
Performance Indicators	19
Content Expansion	19
Application of Knowledge, Skills and Values (AKSA)	19

SECRETARY'S MESSAGE

This Biology Teacher Resource Book is to be used exclusively by teachers of Biology to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Biology Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Biology before undertaking the advanced concepts introduced in this Teacher Resource Book.

The Teacher Resource Book conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Biology as a fundamental field of science.

This STEM Biology Teacher Resource Book for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Biology.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Biology is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Biology to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this Teacher Resource Book is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this Teacher Resource Book as the official curriculum for Biology to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Biology is the study of living organisms and how they interact at multiple levels to sustain life. The basic building block of life is considered to be the cell. The hereditary units are the genetic sequences and these are encased within the cells. This genetic information ensures the hereditary traits necessary for survival of the species is passed on from parents to progenies. Biology is divided into different fields of study. For instance, botany is the study of plants whilst zoology involves the study of animals. Micro-organisms are studied in microbiology. Recent fields that emerged in the last few decades include molecular biology, genomics, and so on. Since Biology is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Biology. Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills. The modules in this Teacher Resource Book are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

The Grade 11 Biology Teacher Resource Book entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Teacher Resource Book are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

THEMA	MODULE	TOPICS
BIODIVERSITY	Agricultural Biodiversity	1.Agricultural Biodiversity in Papua New Guinea 2.Genetic Diversity
	Ornithology	1. Birds of Paradise 2. Inspirations from bird behaviour
BIOTECHNOLOGY	Concepts and Applications of Biotechnology	1.Recombinant DNA Technology 2.Industrial Biotechnology
HEALTH AND MEDICINE	Medicine and Health Sciences	1.Pharmacology 2.Immunization and Vaccines 3.Public Health
ENERGY	Biological Energy Systems	1. General Energy 2. Energy in Animal Cells 3. Energy in Plant Cells 4. Energy and environment
BATTERY	Biological Battery	1. Concepts of Battery

TECHNOLOGY	Systems	2. Biological Inspirations of Battery technology 3. Bio-Batteries
EARTH SCIENCE	Environment	
WASTE MANAGEMENT	Biological Wastes	
INFRASTRUCTURE	Natural Inspirations in Infrastructure	
	Agricultural Infrastructures	
INFORMATION SYSTEM		1.
ARTIFICIAL INTELLIGENCE	Biological Concepts in Artificial Intelligence	1. Neurobiology 2. Neurobiological models in Artificial Intelligence
ENTERPRENEURSHIP		1. Intellectual Property (Patents, Copyrights, Trademarks, etc) 2. How to protect Intellectual Property 3. How to commercialize Intellectual Property

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary.

In the Biology Teacher Resource Book students were introduced to the basic concepts and knowledge of Biology in order for them to develop an appreciation of the discipline in a broader context. In this Teacher Resource Book students are introduced to advanced concepts in Biology as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. This Teacher Resource Book will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM BIOLOGY CONTENT OVERVIEW – TERM 3

Table 1.1: Thema Sequence and Duration

Grade 11 Term 2: STEM-proper Course		
ENERGY (5 Weeks)	1. Biological Energy Systems	1. General Energy Types 2. Energy in Animal Cells 3. Energy in Plant Cells 4. Energy and Environment
BATTERY TECHNOLOGY (3 Weeks)	1. Biological Battery Systems	1. Concepts of Battery 2. Biological Inspirations of Battery Technology 3. Bio-Batteries

Table 1.2 Sequencing & Benchmarking STEM Biology Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Energy	Broad Content Standard	
Module 1	Biological Energy Systems	11.1: Appreciate and describe the biological systems that generate, store, utilize and conserve energy.	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. General Energy Types	11.1.1a: Define energy generally and identify the varying energy types. 11.1.1b: Describe potential, kinetic and electrical energy in biological terms or context.	Students master the ability and knowledge to: - Identify and describe the varying biological energy systems - Describe the energy generation, storage and utilization mechanisms in animal cells - Describe the energy generation, storage and utilization mechanisms in plant cells - Describe the global impacts of energy wastes and strategies to mitigate environmental stress	Research Project + Written Test
2. Energy in Animal Cells	11.1.2a: Describe the chemical energy generated, stored and utilized in human cells.		
3. Energy in Plant Cells	11.1.1a: Describe the solar energy that is utilized in the process of photosynthesis to produce energy in plants that is stored and utilized as food for animals.		

4. Energy and Environment	11.1.1a: State global warming and the need to minimise CO ₂ emission using terrestrial plants and microalgae to harness solar energy, utilise CO ₂ and produce biofuels		
Thema	Battery Technology	Broad Content Standard	
Module 1	Biological Battery Systems	11.2: Appreciate the basic concepts of batteries to describe the biological battery systems that inspire innovations for future battery technology.	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. Concepts of Battery	11.1.1a: Describe the basic concepts behind battery systems 11.1.1b: Confer the need for alternate sources of energy	Students' ability to demonstrate knowledge and skills to; - Describe the basic concepts of batteries - Describe the biological inspirations to battery technology - Describe the principle concept and applications of bio-batteries	Research Project + Written Test
2. Biological Inspirations of Battery Technology	11.1.1a: Describe battery technology that is resembled and inspired in biological systems.		
3. Bio-Batteries	11.1.1a: Describe the concept of bio-batteries and its applications.		

Table 1.3 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	11.1	11.2
Energy Biological Energy Systems	Appreciate and describe the biological systems that generate, store, utilize and conserve energy.	✓	
Battery Technology Biological Battery Systems	Appreciate the basic concepts of batteries to describe the biological battery systems that inspire innovations for future battery technology.		✓

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THEMA: ENERGY

MODULE: BIOLOGICAL ENERGY SYSTEMS

Duration: 5 Weeks

Linking Previous Module

STEM Bridging;

- Module 2: Cells
- Module 3: Metabolism
- Module 4: Anatomy and Physiology
- Module 6: Ecology

STEM Proper-Term 2;

- Thema 1: Biodiversity
- Thema 2: Biotechnology

Module Broad Content Standard

- Appreciate and describe the biological systems that generate, store, utilize and conserve energy.

Module Content Benchmark

Students master the ability and knowledge to:

- Identify and describe the varying biological energy systems
- Describe the energy generation, storage and utilization mechanisms in animal cells
- Describe the energy generation, storage and utilization mechanisms in plant cells
- Describe the global impacts of energy wastes and strategies to mitigate environmental stress.

Topic 1: General Energy Types

Performance Indicators

11.1.1a: Define energy generally and identify the varying energy types.

11.1.1b: Describe potential, kinetic and electrical energy in biological terms or context.

Content Expansion

Body of Knowledge for their understanding

- Kinetic energy
- Potential energy
- Energy conversions
- Electrical Energy and Conduct of Nerve Impulse
 - Action Potential & Nerve Impulse
 - Transmitting Nerve Impulses From One Neuron To Another

Set of Skills for their proficiency levels

- Describe and link kinetic, potential and electrical energy system concepts in biology.

Trait of Attitudes and Values to govern their lives

- Appreciate kinetic energy as the energy of motion
- Appreciate potential energy as the energy of position or structure
- Appreciate electrical energy and the conduction of action potentials and nerve impulse.

Topic 2: Energy in Animal Cells

Performance Indicators

11.1.1a: Describe the chemical energy generated, stored and utilized in human cells.

Content Expansion

Body of Knowledge for their understanding

- Chemical Energy and Cellular Respiration
- Electrons and Energy
- Electron carriers
- Adenosine Triphosphate (ATP) in Living Systems
 - ATP Structure and Function
 - Energy from ATP
- Phosphorylation
- Outcomes of Glycolysis

Set of Skills for their proficiency levels

- Describe the process of energy (ATP) generation in cells
- Confer the similarities of electron transport chain in animal and plant cells.
- Describe the outcomes of cellular respiration via glycolysis

Trait of Attitudes and Values to govern their lives

- Appreciate the concepts surrounding electron transfer in cellular respiration for production of energy as ATP.
- Appreciate how energy is being generated, stored and utilized in animal cells.

Topic 3: Energy in Plant Cells

Performance Indicators

11.1.1a: Describe the solar energy that is utilized in the process of photosynthesis to produce energy in plants that is stored and utilized as food for animals.

Content Expansion

Body of Knowledge for their understanding

- Photosynthesis and the Flow of Solar Energy to Living Organisms
- Evolution Of The Photosynthetic Process
- Light & Dark Reactions Of Photosynthesis (The Strategy)
- Lessons Learnt From The Process Of Photosynthesis (The Potential)

Set of Skills for their proficiency levels

- Describe the energy from the sun that is key to photosynthetic processes in plant cells.
- Confer the similarities of electron transport chain in animal and plant cells.

Trait of Attitudes and Values to govern their lives

- Appreciate the flow of solar energy within living organisms
- Appreciate the significance of the process of photosynthesis on life forms

Topic 4: Energy and Environment

Performance Indicators

11.1.1a: State global warming and the need to minimise CO₂ emission using terrestrial plants and microalgae to harness solar energy, utilise CO₂ and produce biofuels

Content Expansion

Body of Knowledge for their understanding

- Global Warming – A Real Life Situation
- Biological energy production
 - Energy and environmental issues

- Photosynthesis and biomass
 - Photosynthetic efficiency
- Biomass wastes and their conversion to generate methane and ethanol
- General problems

Set of Skills for their proficiency levels

- Discuss the negative impacts of wasted energy resulting in Global Warming.
- Describe the importance of mitigating global warming by means of promoting renewable energy sources as in utilizing our biomass.

Trait of Attitudes and Values to govern their lives

- Appreciate the issues associated with negative impacts of energy as in Global Warming and Green House Effect.
- Appreciate the immense biomass that Papua New Guinea has and how to utilize it for clean energy generation purposes.

Application of Knowledge, Skills and Values (AKSA)

1. Appreciate the Significance of energy that can never be discreet as it is universally applicable in all aspects of life.
2. Students gain the ability to aware of energy in varying forms and focus on possible ways to harness and utilize it for production or industrial use.

THEMA: BATTERY TECHNOLOGY

MODULE: BIOLOGICAL BATTERY SYSTEMS

Duration: 5 Weeks

Linking Previous Module

STEM Bridging;

- Module 2: Cells
- Module 3: Metabolism
- Module 4: Anatomy and Physiology
- Module 6: Ecology

STEM Proper-Term 2;

- Thema 1: Biodiversity
- Thema 2: Biotechnology

Module Broad Content Standard

- Appreciate the basic concepts of batteries to describe the biological battery systems that inspire innovations for future battery technology.

Module Content Benchmark

Students master the ability and knowledge to:

- Describe the basic concepts of batteries
- Describe the biological inspirations to battery technology
- Describe the principle concept and applications of bio-batteries

Topic 1: Concepts of Battery

Performance Indicators

11.1.1a: Describe the basic concepts behind battery systems

11.1.1b: Confer the need for alternate sources of energy

Content Expansion

Body of Knowledge for their understanding

- What is a battery?
- Common types of batteries
 - Primary Batteries
 - Secondary Batteries
- How do (rechargeable) batteries work?
- The Need For Alternative Battery Sources

Set of Skills for their proficiency levels

- Describe the underlying concepts of batteries
- Discuss the different types of batteries, how they work and why we need alternate sources of battery.

Trait of Attitudes and Values to govern their lives

- Appreciate batteries as a chemical energy storage medium that can generate electrical potential energy to do work.

Topic 2: Biological Inspirations of Battery Technology

Performance Indicators

11.1.1a: Describe battery technology that is resembled and inspired in biological systems.

Content Expansion

Body of Knowledge for their understanding

- Battery Systems Of The Human Body
- Generation and conduct of nerve impulse
 - Giant Squid
- Initiation of Heart Beat: Role of the Pacemaker
- The Electric Eel

Set of Skills for their proficiency levels

- Describe the battery system that powers the human body
- Discuss the electrical energy system resembling battery technology in conduction of nerve impulse (as in Giant Squid), the initiation of heartbeat (Pacemaker) and the Electric Eel.

Trait of Attitudes and Values to govern their lives

- Appreciate the battery systems in living organisms (humans, giant squid, electric eel, etc.) that inspire existing technologies as well as future innovations.
-

Topic 3: Bio-Batteries

Performance Indicators

11.1.1a: Describe the concept of bio-batteries and its applications.

Content Expansion

Body of Knowledge for their understanding

- What is bio-battery?
- Bio-battery working principle
- Types of bio-batteries
- Advantages of bio-battery
- Disadvantages of bio-battery
- Applications of bio-battery
- Transforming The Body's Thermal Heat Into Human Battery

Set of Skills for their proficiency levels

- Define bio-batteries and describe their working principle
- Confer the types, pros and cons of bio-batteries

Trait of Attitudes and Values to govern their lives

- Appreciate the significance of bio-batteries
- Appreciate the applications of bio-batteries
- Appreciate battery technology as a topical and potential field of innovation in the near future.

Application of Knowledge, Skills and Values (AKSA)

3. Appreciate battery technology as the potential area where the world will resort to in the near future as there will be a need to store energy in batteries for convenient, efficient and effective use.
4. The ability of students to realize the future need for battery technology and from biological inspirations they can innovate.

